

Name:

Student ID:

VECTOR SPACE PROBLEM SHEET

1

- (a) Show that if the columns, $\mathbf{m}_j$, of an $n \times n$ matrix, $\mathbf{M}$, are not independent then $\mathbf{M}$ does not have a unique inverse. For the columns to not be independent implies there exists some set of coefficients c_j such that

$$\sum_{j=1}^n c_j \mathbf{m}_j = \mathbf{0}.$$

We note that the i^{th} element of $\mathbf{m}_j = M_{ij}$ so

$$\sum_{j=1}^n c_j M_{ij} = 0$$

[3 marks]

3

- (b) Show that for three $n \times n$ matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ the inverse of $\mathbf{ABC}$ is $\mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$. You can do this by multiplying the product of matrices, but its inverse and use associativity of matrix multiplication [3 marks]

3

- (c) For non-square matrices, **A** and **B** we can have $\mathbf{AB} = \mathbf{I}$ while $\mathbf{BA} \neq \mathbf{I}$. Show this is the case for the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \end{pmatrix} \qquad \mathbf{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

[3 marks]

3

- (d) For finite-dimensional square matrices if $\mathbf{AB} = \mathbf{I}$ then $\mathbf{BA} = \mathbf{I}$. Thus $\mathbf{B} = \mathbf{A}^{-1}$. This is not entirely trivial to prove. We break the proof down. Show that if $\mathbf{AB} = \mathbf{I}$ then **B** is injective (one-to-one). This means that for two different vectors x and y then $\mathbf{B}x \neq \mathbf{B}y$. Equivalently $\mathbf{B}(x - y) \neq 0$, unless $x = y$, but as x and y are arbitrary the only vector for which $\mathbf{B}x = 0$ is the vector $x = 0$. Prove that if $\mathbf{AB} = \mathbf{I}$ and $\mathbf{B}x = 0$ then $x = 0$. [2 marks]

2

- (e) Next we want to show that in a finite dimensional space a square matrix that is injective is also surjective (onto). That is, if $\mathbf{B}$ is injective then for any vector $\mathbf{y}$ there exists a vector $\mathbf{x}$ such that $\mathbf{y} = \mathbf{B}\mathbf{x}$. The reason a finite dimensional square matrix that is injective is also surjective is not so obvious. To prove this we consider a set of n **basis** vectors that span the entire space $\mathbb{R}^n$. We call these $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$. Because these are independent the only set of numbers c_i such that $\sum_i c_i \mathbf{v}_i = \mathbf{0}$ are when $c_1 = c_2 = \dots = c_n = 0$. Show that the set of vectors $\mathbf{v}'_i = \mathbf{B}\mathbf{v}_i$ form a set of basis vectors (i.e. they are linear independent). This is easily proved by contradiction. [5 marks]

5

- (f) Use the fact that the vectors $\mathbf{v}'_i$ are linearly independent to show that, for any vector $\mathbf{y}$, we can find a vector $\mathbf{x}$ such that $\mathbf{y} = \mathbf{B}\mathbf{x}$ [2 marks]

2

(g) Using the fact that $\mathbf{AB} = \mathbf{I}$ and $\mathbf{B}$ is surjective show $\mathbf{BA} = \mathbf{I}$ [2 marks]

2

End of question 1

(a) $\frac{1}{3}$	(b) $\frac{1}{3}$	(c) $\frac{1}{3}$	(d) $\frac{1}{2}$	(e) $\frac{1}{5}$	(f) $\frac{1}{2}$	(g) $\frac{1}{2}$	Total $\frac{1}{20}$
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2 Throughout this question you may use two standard properties of the trace, namely that a matrix and its transpose have the same trace, and that the trace is unchanged by cyclically permuting a product

$$\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}^T)$$

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA}).$$

(a) For a vector $\mathbf{a} \in \mathbb{R}^m$, a vector $\mathbf{b} \in \mathbb{R}^n$ and an $m \times n$ matrix $\mathbf{U}$, show that

$$\mathbf{a}^T \mathbf{U} \mathbf{b} = \text{tr}(\mathbf{U}^T \mathbf{a} \mathbf{b}^T).$$

The quickest route is to write both sides out in components. Note that $\mathbf{a} \mathbf{b}^T$ is an $m \times n$ matrix while $\mathbf{a}^T \mathbf{U} \mathbf{b}$ is a scalar. [5 marks]

(b) Verify the identity explicitly for

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \mathbf{U} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

by computing both $\mathbf{a}^T \mathbf{U} \mathbf{b}$ and $\text{tr}(\mathbf{U}^T \mathbf{a} \mathbf{b}^T)$. [4 marks]

$\frac{1}{4}$

- (c) For a square matrix $\mathbf{U}$ and a sufficiently small scalar ϵ , the inverse of $\mathbf{I} + \epsilon\mathbf{U}$ is given by the geometric (Neumann) series

$$(\mathbf{I} + \epsilon \mathbf{U})^{-1} = \mathbf{I} - \epsilon \mathbf{U} + \epsilon^2 \mathbf{U}^2 - \epsilon^3 \mathbf{U}^3 + \dots.$$

Show this by multiplying $\mathbf{I} + \epsilon \mathbf{U}$ by the truncated series

$$\mathbf{s}_n = \sum_{k=0}^n (-\epsilon)^k \mathbf{u}^k$$

and showing that the result differs from $\mathbf{I}$ only at order ϵ^{n+1} . Hence write down $(\mathbf{I} + \epsilon \mathbf{U})^{-1}$ to first order in ϵ . [5 marks]

[illegible]

- (d) Use this to find the derivative of $g(\mathbf{X}) = \mathbf{a}^\top \mathbf{X}^{-1} \mathbf{b}$ with respect to the matrix $\mathbf{X}$. Replace $\mathbf{X}$ by $\mathbf{X} + \epsilon \mathbf{U}$, expand to first order in ϵ , and use the definition of the derivative with respect to a matrix

$$\lim_{\epsilon \rightarrow 0} \frac{g(\mathbf{X} + \epsilon \mathbf{U}) - g(\mathbf{X})}{\epsilon} = \text{tr} \left(\mathbf{U}^\top \frac{\partial g(\mathbf{X})}{\partial \mathbf{X}} \right)$$

which holds for every $\mathbf{U}$. Part (a) is what lets you read the derivative off.

[6 marks]

6

End of question 2

(a) $\frac{5}{5}$	(b) $\frac{4}{4}$	(c) $\frac{5}{5}$	(d) $\frac{6}{6}$	Total $\frac{20}{20}$
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3 We want to compute the derivative of $\log(\det(\mathbf{X}))$ with respect to the matrix $\mathbf{X}$. Our strategy will be to use the fact that

$$\lim_{\epsilon \rightarrow 0} \frac{\log(\det(\mathbf{X} + \epsilon \mathbf{U})) - \log(\det(\mathbf{X}))}{\epsilon} = \text{tr} \mathbf{U}^T \frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}.$$

We therefore expand $\log(\det(\mathbf{X} + \epsilon \mathbf{U}))$ to first order in ϵ , subtract $\log(\det(\mathbf{X}))$, divide by ϵ , rearrange and read off $\frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}$. However, rearranging in this case is quite complicated.

- (a) Using $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$ and $\log(ab) = \log(a) + \log(b)$ expand $\log(\det(\mathbf{X} + \epsilon \mathbf{U}))$ as a sum of $\log(\det(\mathbf{X}))$ and another logarithm. [5 marks]

5

- (b) The hard part is to expand $\det(\mathbf{I} + \epsilon \mathbf{M})$. Write out $\det(\mathbf{I} + \epsilon \mathbf{M})$ explicitly for a 2 dimensional matrix and expand to first order in ϵ [2 marks]

2

(c) For an $n \times n$ matrix $\mathbf{M}$ compute the determinant of

$$\det(\mathbf{I} + \epsilon \mathbf{M}) = \det \begin{pmatrix} 1 + \epsilon M_{11} & \epsilon M_{12} & \dots & \epsilon M_{1n} \\ \epsilon M_{21} & 1 + \epsilon M_{22} & \dots & \epsilon M_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \epsilon M_{n,1} & \epsilon M_{n2} & \dots & 1 + \epsilon M_{nn} \end{pmatrix}$$

up to first order in ϵ . Recall we can compute the determinant a matrix $\mathbf{X}$ as

$$\det(\mathbf{X}) = \sum_{i=1}^n (-1)^j X_{1j} \det(\hat{\mathbf{X}}_{1j})$$

where $\hat{\mathbf{X}}_{1j}$ is the matrix obtained by removing row 1 and column j from the matrix $\mathbf{X}$. [5 marks]

5

(d) For a more elegant proof we can use the fact that

$$\det(\mathbf{M}) = \prod_{i=1}^n \lambda_i \qquad \text{tr} \mathbf{M} = \sum_{i=1}^n \lambda_i$$

where λ_i are the eigenvalues of the matrix $\mathbf{M}$. We will prove these results later when we look at eigen-systems. Note that if $\mathbf{v}_i$ is an eigenvector of the matrix $\mathbf{M}$, with eigenvalue λ_i then it is an eigenvector of $\mathbf{I} + c\mathbf{M}$ with eigenvalue $1 + c\lambda_i$ since

$$(\mathbf{I} + c\mathbf{M})\mathbf{v}_i = \mathbf{v}_i + c\lambda_i\mathbf{v}_i = (1 + c\lambda_i)\mathbf{v}_i.$$

Using these results show that $\det(\mathbf{I} + \epsilon\mathbf{M}) = 1 + \epsilon\text{tr} \mathbf{M} + O(\epsilon^2)$. [3 marks]

3

(e) Using the results

$$\begin{aligned} \log(\det(\mathbf{X} + \epsilon\mathbf{U})) - \log(\det(\mathbf{X})) &= \log(\det(\mathbf{I} + \epsilon\mathbf{X}^{-1}\mathbf{U})) \\ \det(\mathbf{I} + \epsilon\mathbf{M}) &= 1 + \epsilon\text{tr} \mathbf{M} + O(\epsilon^2) \end{aligned}$$

together with the fact that $\log(1 + x) = x + O(x^2)$, compute $\frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}$. You may use the fact that $\log(1 + x) = x + O(x^2)$ and $\text{tr} \mathbf{A} = \text{tr} \mathbf{A}^T$. [5 marks]

5

End of question 3

(a) $\frac{1}{5}$ (b) $\frac{1}{2}$ (c) $\frac{1}{5}$ (d) $\frac{1}{3}$ (e) $\frac{1}{5}$ Total $\frac{1}{20}$